

CODE:

```
#include<stdio.h>
#include<windows.h>

#define N 5

void philosopher(int id){
    printf("Philosopher %d is thinking\n", id);
    Sleep(1000);

    printf("Philosopher %d picked left chopstick\n", id);
    printf("Philosopher %d picked right chopstick\n", id);

    printf("Philosopher %d is eating\n", id);
    Sleep(1000);

    printf("Philosopher %d finished eating\n\n", id);
}

int main(){
    int i;
    for(i=0;i<N;i++){
        philosopher(i);
    }
    return 0;
}
```

OUTPUT:

```
PS C:\OSLab_CM24118\dining.c> gcc dining.c -o dining
PS C:\OSLab_CM24118\dining.c> .\dining
Philosopher 0 is thinking
Philosopher 0 picked left chopstick
Philosopher 0 picked right chopstick
Philosopher 0 is eating
Philosopher 0 finished eating

Philosopher 1 is thinking
Philosopher 1 picked left chopstick
Philosopher 1 picked right chopstick
Philosopher 1 is eating
Philosopher 1 finished eating

Philosopher 2 is thinking
Philosopher 2 picked left chopstick
Philosopher 2 picked right chopstick
Philosopher 2 is eating
Philosopher 2 finished eating

Philosopher 3 is thinking
Philosopher 3 picked left chopstick
Philosopher 3 picked right chopstick
Philosopher 3 is eating
Philosopher 3 finished eating

Philosopher 4 is thinking
Philosopher 4 picked left chopstick
Philosopher 4 picked right chopstick
Philosopher 4 is eating
Philosopher 4 finished eating
```